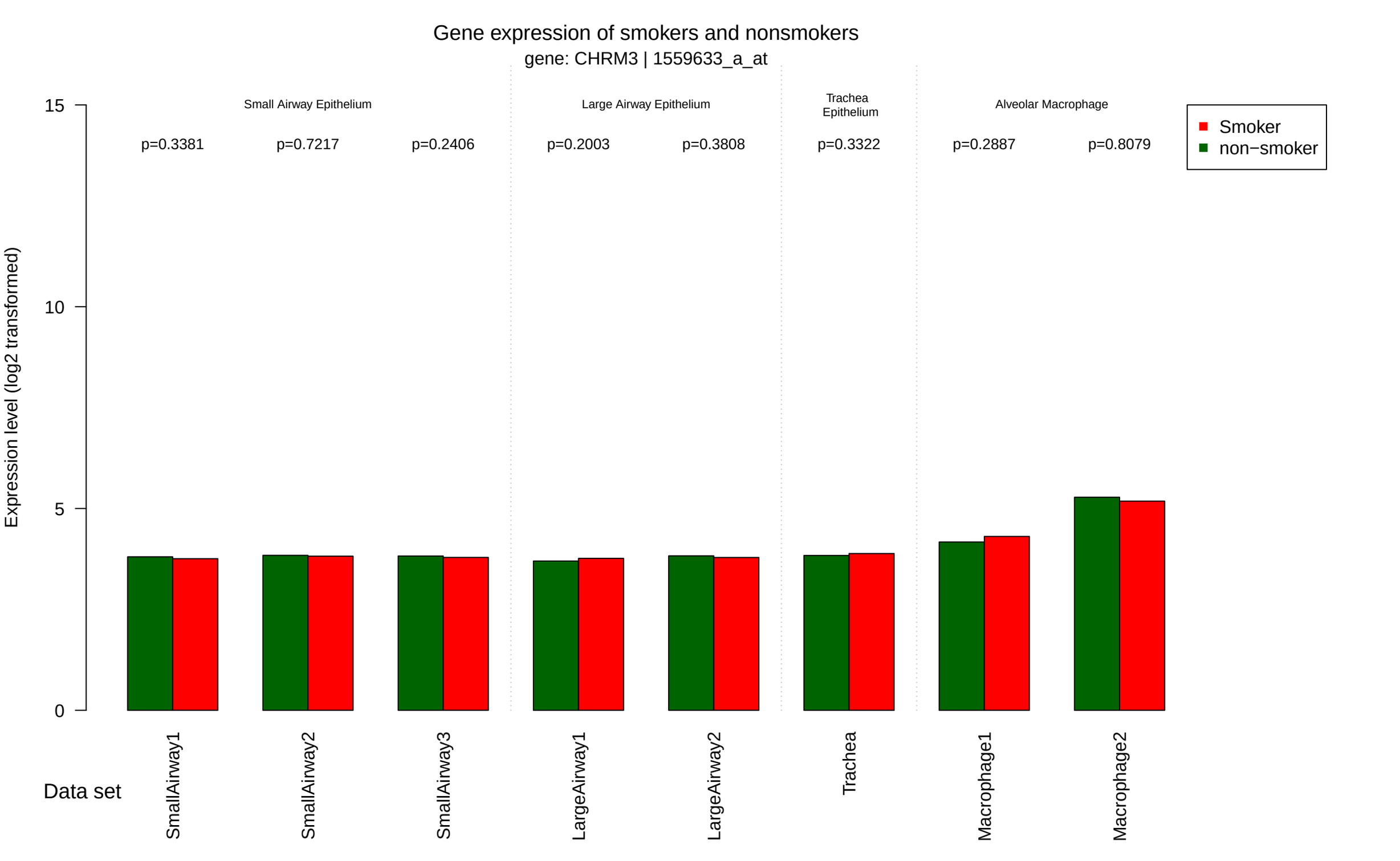
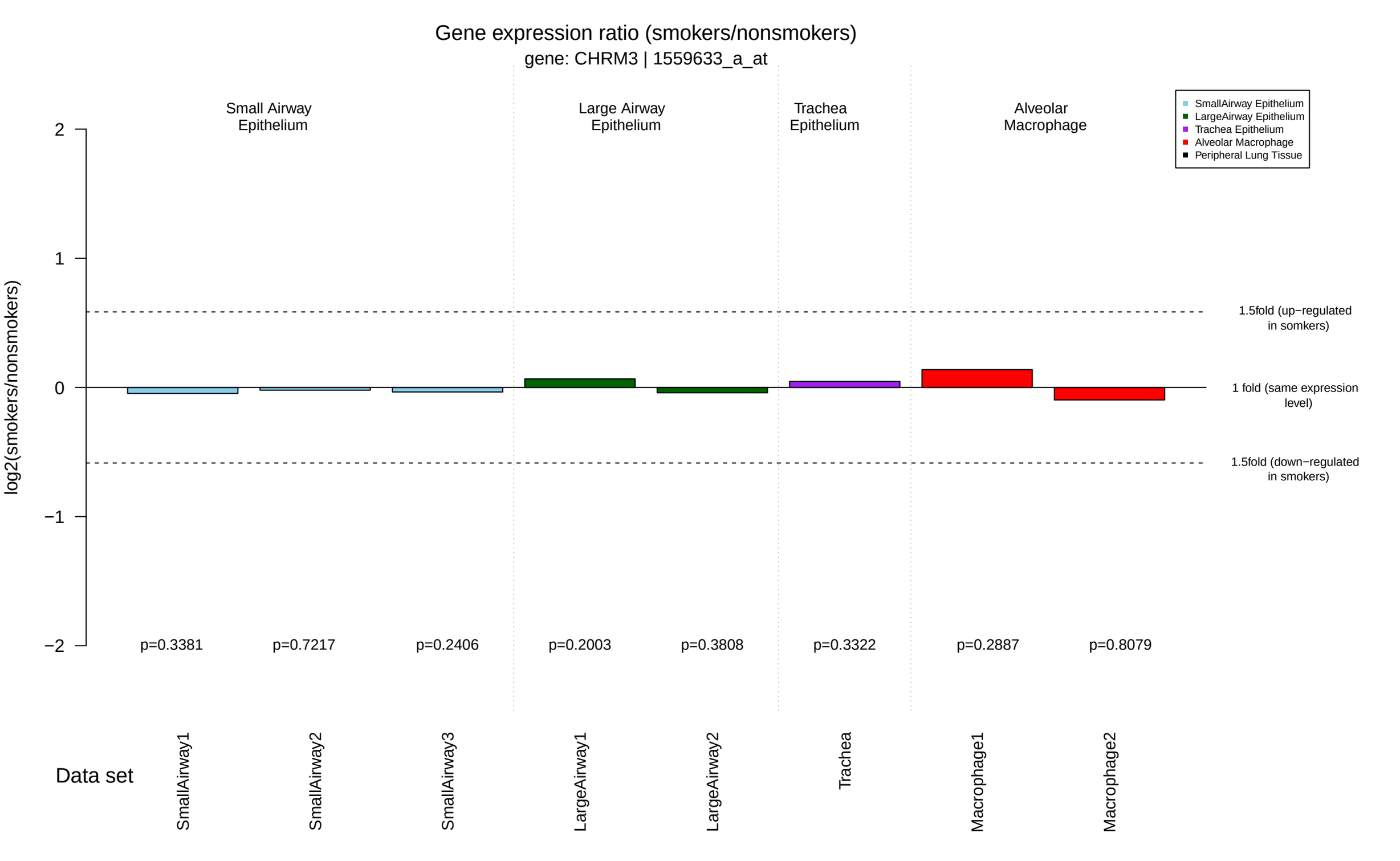
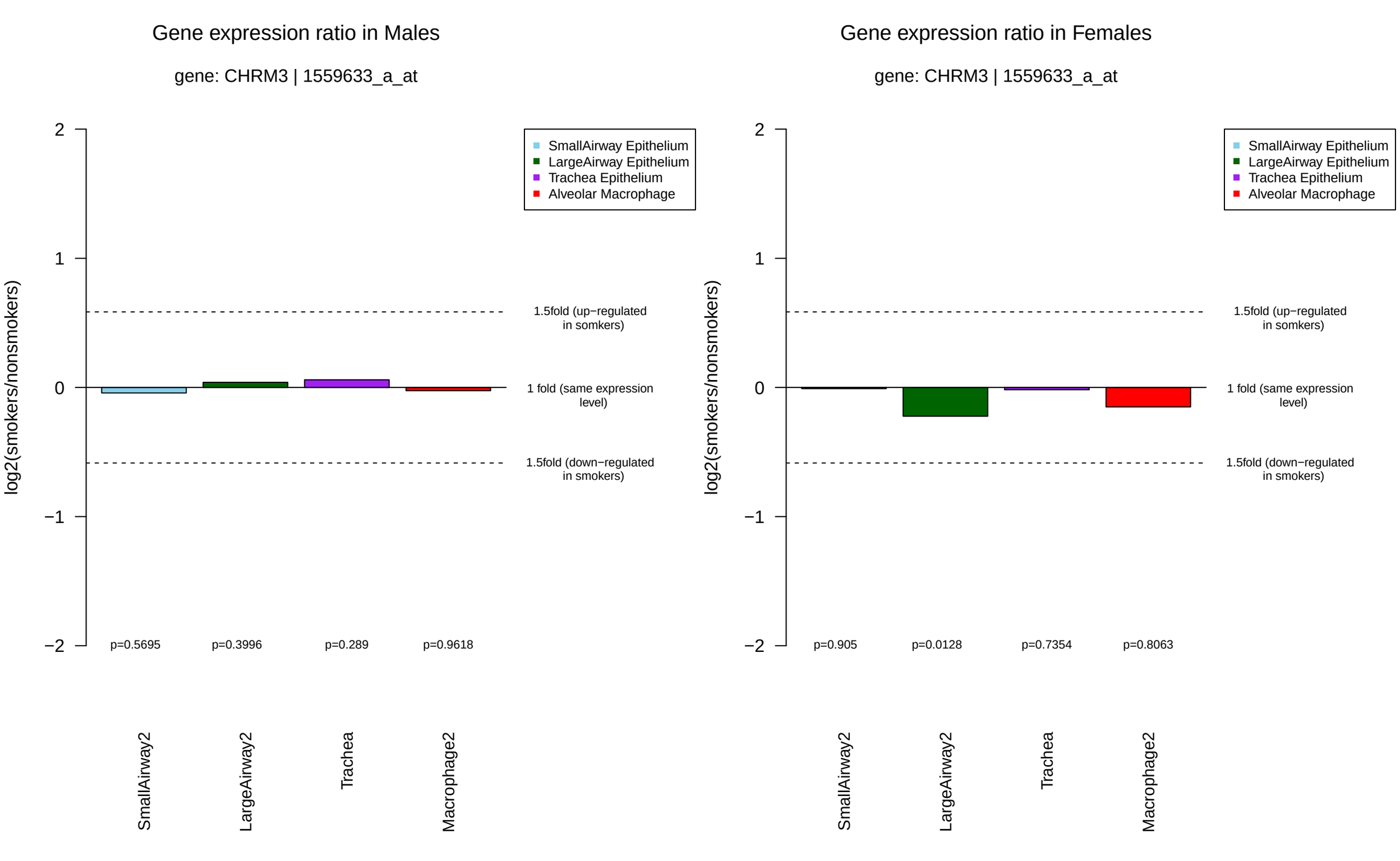
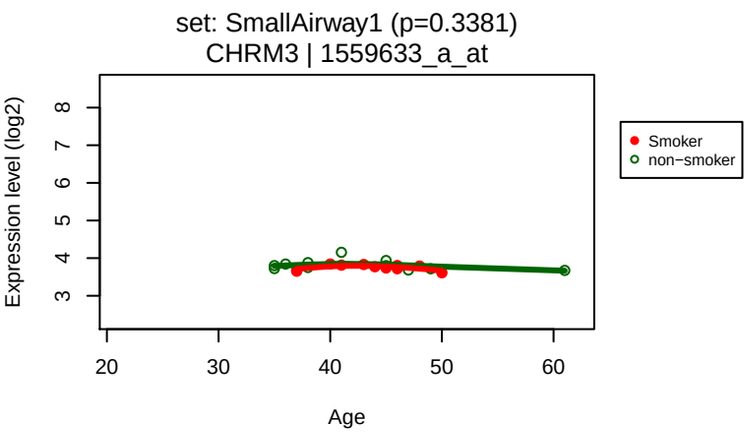
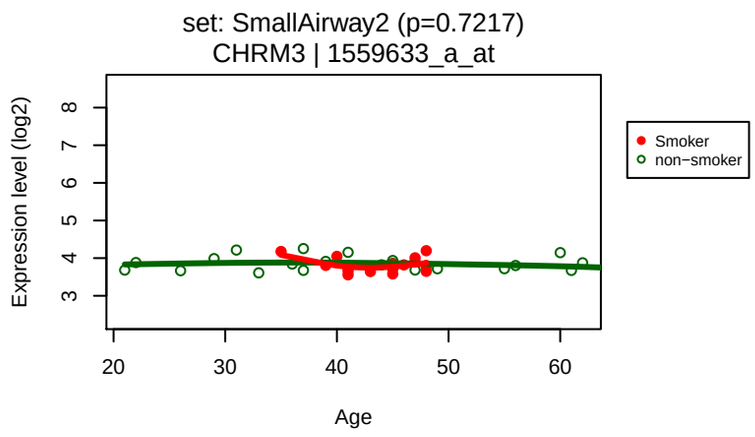
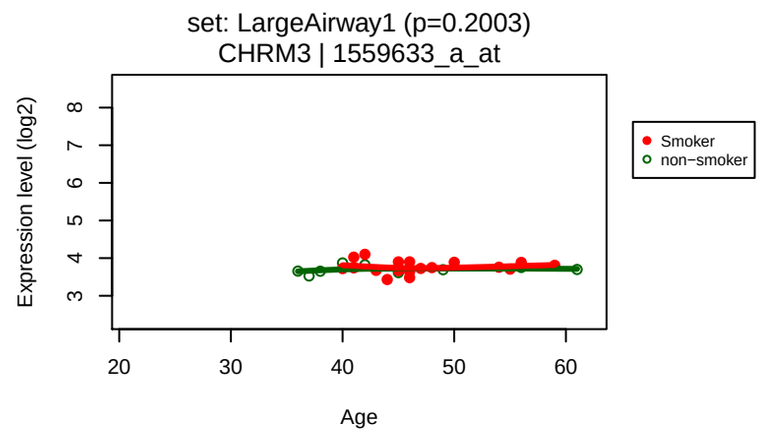

Gene expression of smokers and nonsmokers
gene: CHRM3 | 1559633_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3381
p=0.7217
p=0.2406
p=0.2003
p=0.3808
p=0.3322
p=0.2887
p=0.8079
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

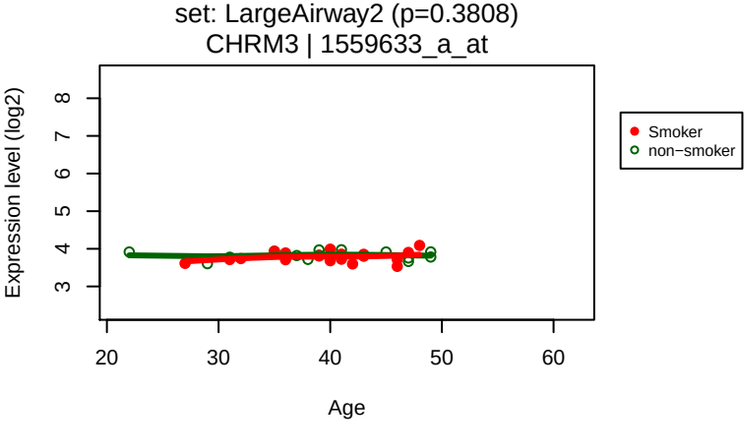

Gene expression ratio (smokers/nonsmokers)
gene: CHRM3 | 1559633_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3381
p=0.7217
p=0.2406
p=0.2003
p=0.3808
p=0.3322
p=0.2887
p=0.8079
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

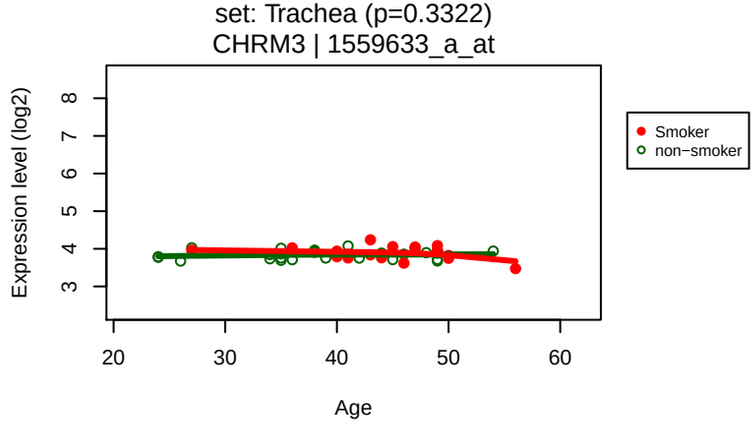

Gene expression ratio in Males
gene: CHRM3 | 1559633_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5695
p=0.3996
p=0.289
p=0.9618
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM3 | 1559633_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.905
p=0.0128
p=0.7354
p=0.8063
SmallAirway2
LargeAirway2
Trachea
Macrophage2

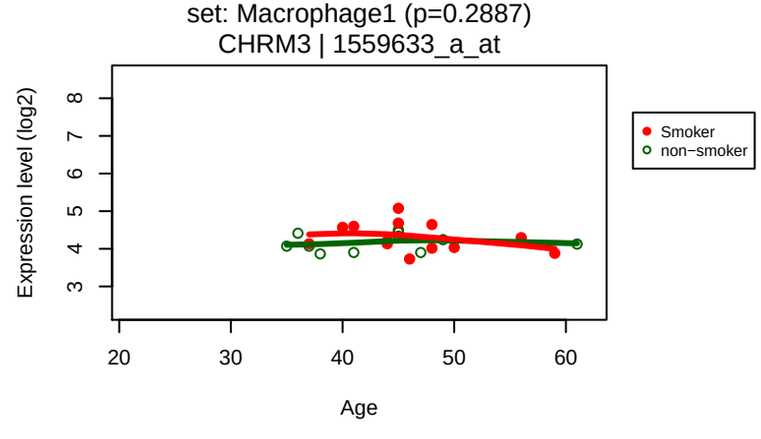

set: SmallAirway1 (p=0.3381)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker

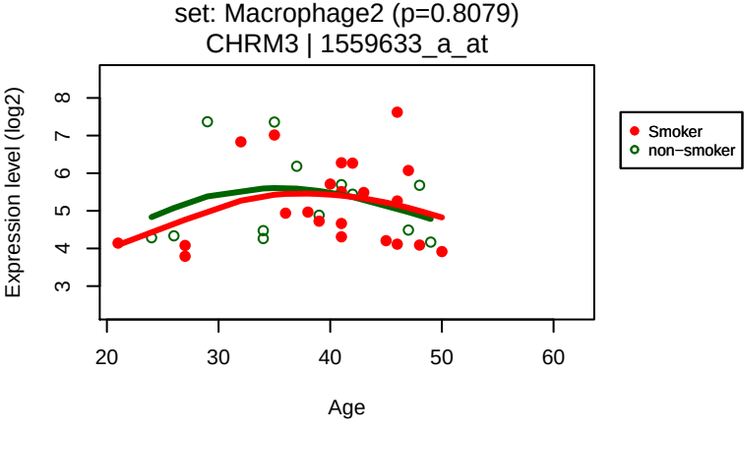

set: SmallAirway2 (p=0.7217)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker

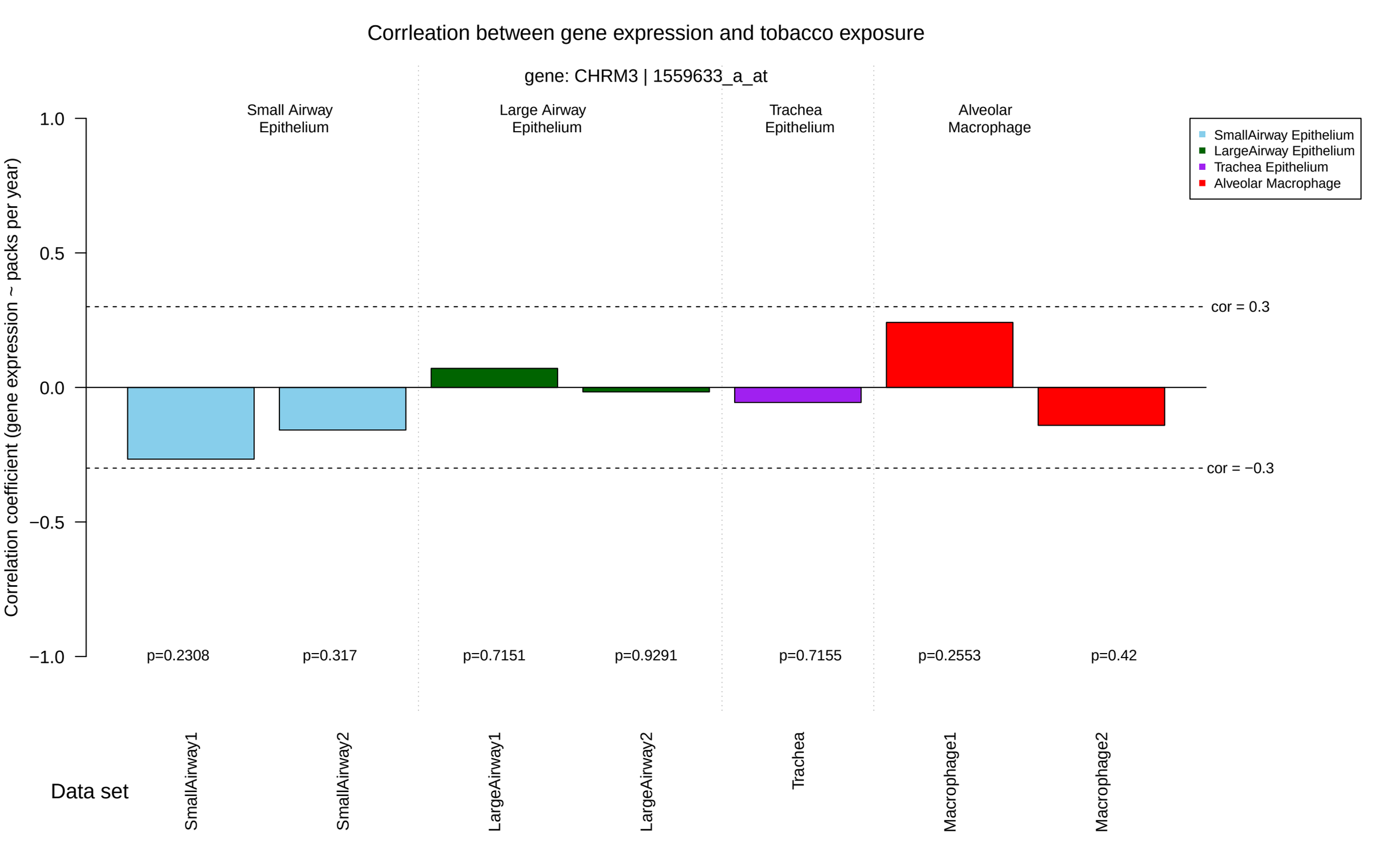

set: LargeAirway1 (p=0.2003)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3808)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3322)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2887)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8079)
CHRM3 | 1559633_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM3 | 1559633_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2308
p=0.317
p=0.7151
p=0.9291
p=0.7155
p=0.2553
p=0.42
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2